UNIVERSAL MODULAR PLC Hardware Requirement Specification (HRS)

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Prepared for: Hardware Engineering Team
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1. SYSTEM OVERVIEW

The Universal PLC system shall be a modular industrial controller designed for automation, data acquisition, and IoT connectivity.
Key objectives:

- Industrial grade PLC controller
- Support industrial communication protocols (Modbus Serial, Modbus TCP/IP)
- Support Open PLC runtime (Ladder / FBD / ST)
- Support modular expansion over I2C/SPI
- Inbuilt Ethernet + RS485 communication
- Optional 4G IoT connectivity with GPS/GNSS
- MQTT cloud integration
- Secure firmware and OTA updates
- Industrial EMC and surge protection
- VPN Facility

Hardware extra features

- RTC with backup battery
- NTP time sync via Ethernet
- GPS time sync via 4G module
- FRAM for data logging
- Supercapacitor for safe shutdown

Operating environment:

Parameter	Requirement
Operating Temperature	0°C to 60°C
Storage Temperature	-20°C to 70°C
Humidity	10–90% non-condensing
Mounting	DIN Rail
Protection Level	Industrial environment

2. SYSTEM ARCHITECTURE

Main modules:

1. PLC CPU Controller Board
2. Digital IO Section
3. Communication Section
4. 4G IoT Gateway Module
5. Energy Meter Module
6. Analog IO Expansion
7. RTD/TC Expansion
8. HMI Interface(UART/Modbus)

Communication architecture

Interface	Usage
Ethernet	Modbus TCP / MQTT/Web UI
RS485	Modbus RTU
I2C	Expansion modules
SPI	Energy meter / SD card
UART	4G module

3. PLC CPU CONTROLLER BOARD

3.1 Microcontroller

Preferred MCU:
STM32F407VGT6

Minimum specifications:

Parameter	Requirement
CPU Frequency	≥168 MHz
Flash	≥1MB
RAM	≥192 KB
Timers	≥10
DMA	Required
Ethernet MAC	Required
SPI	≥3
I2C	≥2
UART	≥4

3.2 EXTERNAL MEMORY

Component	Requirement
SPI Flash	16MB minimum
SD Card	MicroSD support
File System	FAT32

Used for:

- PLC program storage
- Data logging
- Firmware updates
- OTA backup

4. POWER SUPPLY

Input Voltage: 12–24V DC
Protection requirements:

Feature	Requirement
Reverse polarity protection	Mandatory
Input surge protection	Mandatory
Overvoltage protection	Required
EMI filter	Required

Recommended components:

- TVS diode
- LC filter
- Reverse MOSFET protection

5. DIGITAL INPUT SECTION

Total Inputs: 18
Configuration:

Group 1 : DI1–DI6
Group 2 : DI7–DI12
Group 3 : DI13–DI18

Each group shall have configurable common terminal.

Input characteristics:

Parameter	Requirement
Input voltage	5–24V DC
Logic type	PNP / NPN selectable
Isolation	Optocoupler
Input current	3–8mA

5.1 HIGH SPEED INPUT

- First 6 inputs support encoder pulses.
- Maximum frequency: 100kHz
- Signal Type: 5-24V DC

Recommended optocouplers: 6N137 HCPL2631 ACPL-M61L
Standard optocouplers like PC817 must not be used for high-speed inputs.

6. DIGITAL OUTPUT SECTION

Total Outputs: 12
Output type: PNP Transistor Output
Output rating: Voltage: 24V DC Current: ≥500mA per channel
Protection: Flyback diode

Recommended driver IC:

- TPS1H200A (Smart High-Side Switch)
- BTS500xx series
- VNQ5050

6.1 HIGH SPEED OUTPUT

First 4 outputs support high speed pulse output.
Maximum frequency: 100kHz Used for stepper/servo control
Outputs must be connected to hardware timer PWM pins.

7. COMMUNICATION INTERFACES

7.1 Ethernet

Parameter	Requirement
Protocol	Modbus TCP
Protocol	MQTT
Protocol	HTTP Web UI
Protocol	OTA Update
IP Mode	Static + DHCP
DNS	Supported

Recommended PHY:

- LAN8720A

Interface:

- RMII

Note:

- Ethernet port should be used for Internet and Modbus TCP/IP Communication at different IP address using sperate socket.

7.2 RS485

Parameter	Requirement
Protocol	Modbus RTU
Mode	Master / Slave
Slave devices	1 to 32

Isolation required.
Recommended components:

- ISO3082 ADM2587E

Protection:

- TVS diode ESD protection

8. EXPANSION BUS

Expansion bus must support external modules.
Supported modules:

- 4G Gateway
- Analog IO
- RTD / TC
- Energy Meter

Bus	Usage
I2C	Analog modules
SPI	Energy meter
UART	4G module

9. 4G IoT + GPS Module

The PLC shall support an optional 4G IoT communication module with integrated GNSS.

Purpose:

- Cellular connectivity
- MQTT cloud communication
- OTA firmware update
- GPS location tracking
- Network time synchronization

Recommended Cellular Module:

- Quectel EC200U (preferred)
- Alternative: Quectel EC25

Features:

- LTE connectivity
- UART interface
- GNSS support (GPS / GLONASS / BeiDou / Galileo)

Hardware Requirements:

- Single
-
- ++++++++++++++ Sim Nano SIM card holder
- LTE antenna connector (U.FL / IPEX)
- GNSS antenna connector
- Power supply capable of 2A peak current

Communication with PLC:

- UART interface

Functions:

- MQTT communication over cellular network
- Ethernet to 4G fallback logic
- OTA firmware download
- GPS location reporting

Signals:

- TX
- RX
- CTS
- RTS
- PWRKEY
- STATUS

10. ENERGY METER MODULE

Metering IC:

- ATM90E36A

Features required:

Measurement	Required
Voltage RMS	Yes
Current RMS	Yes
Power	Yes
Energy	Yes
Harmonics	Yes

Electrical support:

Parameter	Requirement
Voltage	Voltage Range: 350–500V AC Line-Line (3 Phase)
CT	1A / 5A
Accuracy	Class 0.5 / 1

Communication:

- SPI

11. ANALOG IO EXPANSION

Configuration:

- 4 Analog Inputs
- 2 Analog Outputs

Supported signals:

- Voltage: 0–10V
- Current: 0-20mA/4–20mA

ADC requirements: Resolution:

- 12–16 bit DMA sampling

DAC resolution:

- 12–16 bit

Filtering modes:

- Strong
- Medium
- Low

Isolation required.

12. RTD / THERMOCOUPLE MODULE

Supported sensors:

- RTD: 3-wire
- Thermocouple: 2-wire

Recommended IC:

- MAX31865 (RTD)
- MAX31856 (Thermocouple)

13. SD CARD SUPPORT

Functions:

- PLC program upload
- Firmware update
- Data logging

Interface:

- SPI

14. WATCHDOG SYSTEM

Two watchdogs required.

Internal watchdog:

- MCU IWDG

External watchdog

- Mandatory external IC

Recommended IC:

- TPS3430
- MAX706

Purpose:

- Reset system if firmware hangs.

15. DUAL FIRMWARE PARTITION

Flash memory must support two firmware slots.

- Firmware A
- Firmware B

Bootloader selects active firmware.

Purpose:

- Safe OTA updates
- Rollback if update fail

Memory Example

- 0x08000000 Bootloader
- 0x08020000 Firmware A
- 0x08100000 Firmware B

16. SECURE BOOT

Secure boot required to prevent unauthorized firmware.

Requirements:

- Firmware signature verification
- SHA256 hash verification

Recommended crypto library:

- mbedTLS

17. EMC COMPLIANCE

Hardware design must comply with:

Standard	Description
IEC 61000-4-2	ESD
IEC 61000-4-4	EFT
IEC 61000-4-5	Surge

Protection required:

- TVS diodes
- RC filters
- Common mode chokes

18. SURGE PROTECTION

Protection required on:

Interface	Protection
Power Input	TVS diode
RS485	TVS diode
Ethernet	Ethernet surge protector
Digital IO	transient suppression

Recommended:

- SMBJ series TVS

19. PCB DESIGN REQUIREMENTS

Minimum PCB layers: 4

Stackup:

Layer	Function
Top	Signals
Layer 2	Ground
Layer 3	Power
Bottom	Signals

Design requirements:

- separation of analog and digital ground
- short return path
- EMI filtering

20. MECHANICAL DESIGN

Mounting: DIN rail
Terminal connectors: 3.81mm pluggable terminals
Enclosure: Industrial ABS / Polycarbonate

21. EMBEDDED WEB UI OVER ETHERNET

The PLC shall include an **embedded HTTP/HTTPS web server** accessible via the Ethernet port.
Accessible via browser.

The web interface will allow **PLC configuration, monitoring, firmware updates, and program upload** using a standard web browser.

Supported browsers:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox

No external software installation should be required.

21.1 WEB SERVER REQUIREMENTS

Parameter	Requirement
Server Type	Embedded HTTP/HTTPS server
Protocol	HTTP 1.1 / HTTPS
Authentication	Username & Password
Session Management	Required
Max Concurrent Users	≥5
Port	Default 80 / 443

Recommended implementation:

- LWIP HTTP Server
- Embedded REST API

21.2 Web UI Access

User shall access the PLC by entering the IP address: http://PLC_IP_ADDRESS

Example: http://192.168.1.50

If HTTPS enabled: https://192.168.1.50

21.3 USER ACCESS LEVELS

The web UI shall implement role-based access control.

Role	Permissions
Admin	Full configuration access
Engineer	PLC program upload/download
Operator	Monitoring only

21.4 WEB DASHBOARD

Dashboard page must display:

Parameter	Description
PLC Status	Running / Stop / Fault
CPU Load	Processor usage
Memory Usage	RAM / Flash
Ethernet Status	Connected / Disconnected
MQTT Status	Connected / Offline
4G Network Status	Signal strength
PLC Runtime Status	Running / Halt

Real-time IO display updated every second.

21.5 PLC PROGRAM MANAGEMENT

Functions:

- Upload PLC program
- Download PLC program
- Start program
- Stop program
- Restart PLC

Programs stored in SPI flash or SD card.

21.6 PLC PARAMETER CONFIGURATION

Ethernet settings:

- IP address
- Subnet mask
- Gateway
- DNS
- DHCP

MQTT configuration:

- Broker address
- Port Client ID
- Username
- Password

Modbus Master configuration:

- Scan Interval
- Timeout
- Transaction Delay
- 1-32 Transaction Wise
    - RS485 baudrate
    - Parity
    - Slave ID
    - Function Code
    - Start register
    - Data length
    - Write Data Array (if Function code if for Write)

Analog scaling configuration.

21.7 IO MONITORING PAGE

Display:

- Digital Inputs
- Digital Outputs
- Analog Inputs(If Analog Module is Configured and Connected)
- Analog Outputs(If Analog Module is Configured and Connected)
- RTD Inputs(If RTD Module is Configured and Connected)
- TC Inputs(If RTD Module is Configured and Connected)
- Energy Data(If Energy Module is Configured and Connected)

Manual override available for admin users.

21.8 FIRMWARE UPDATE

Firmware update methods:

- Web upload
- SD card
- OTA update

Process:

- Upload firmware
- Checksum verification
- Write to backup partition
- Reboot Activate firmware

21.9 SYSTEM LOGS

Logs must include:

- PLC start
- MQTT connection
- Modbus timeout
- IO module detection
- Energy meter alerts

Log should be Stored in SD card or flash.

21.10 FILE MANAGER

User can:

- Download logs
- Upload configuration
- Upload PLC program
- View data logs

21.11 SECURITY

Feature	Requirement
HTTPS support	Required
Password hashing	SHA256
Session timeout	10 minutes
Firmware signature verification	Required

21.12 REST API

Example endpoints:

- GET /api/status
- GET /api/io
- POST /api/config
- POST /api/restart

Used for:

- SCADA integration
- cloud services
- remote diagnostics

21.13 PERFORMANCE

Parameter	Requirement
Page load time	<2 seconds
IO refresh rate	1 second
Max Web UI users	≥5

21.14 FAIL SAFE BEHAVIOR

If web UI crashes:

- PLC control logic must continue running.
- Web server must run in separate RTOS task.

22. VPN Secure Remote Access

The PLC shall support **secure VPN connectivity** to allow remote engineering access and monitoring.

Purpose:

- Secure remote PLC programming
- Remote HMI access
- Secure cloud connectivity
- Remote diagnostics and troubleshooting

The VPN connection must encrypt all network traffic between the PLC and remote server.

Best options for embedded systems:

- WireGuard
- OpenVPN
- IPSec

The PLC must support two connection methods:

1. **Ethernet VPN**

PLC connects to VPN server via Ethernet.

2. **4G VPN**

PLC connects via cellular network.

The VPN system must support:

Feature	Requirement
Encryption	ChaCha20 / AES
Authentication	Public key
Key exchange	Secure
Connection watchdog	Required
Auto reconnect	Required

23. Test Requirements

Hardware Test Must verify:

Test	Required
Power stability	Yes
Noise immunity	Yes
Temperature test	Yes
EMI test	Yes
Communication stress test	Yes

24. LED Status Block

PLC Board

LED	Meaning
Power	ON
Run	PLC running
Fault	Error
Ethernet Link	Connected
Ethernet Activity	Data transfer
Internet Status	Internet Connected